

Rainwater and Nursery Irrigation

Carpathian OER Commons — Utilities Module

Michel Garand | 2026-10-02 | Version v0.2 | CC BY-SA 4.0

Maturity: L1 — designed and costed, untested. Every cost line is an estimate or a listed price, labelled as such in Section 6; the rainfall, the roof, and the nursery's need are assumed or set until the station record, the walk, and the tree nursery module replace them. Nothing has been built. Read every figure below at this level and no higher.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

Converted from the water working paper, EN v0.1, Sections 6 and 8.5, the last part of that paper still unconverted. Four things change from the paper, decided on 2026-09-29: the runoff factor and the rainfall are now quoted from sources; the nursery's weekly need is read against forestry nursery guidance, which asks more at sowing; the hand pump is priced from a listing; and the case the water module hands over — rainwater taking non-drinking uses when the spring falls between its thresholds — is sized here, not costed.

v0.2, 2026-10-02: the household chose the masonry cistern over the plastic tank. The tank stays in the design as the alternative for a site that needs it. Sections 4, 6, 7, 11, 12, and 13 changed; nothing else.

Permits and local rules. This module is a guide, used in different regions and countries under different laws and guidelines. Depending on where it is built, at what scale, and for whom, permits, registrations, or approvals may be required — for building, water, sanitation, fire, waste, food, or the use of forest and land. Section 10 records what was found for Ukraine, dated, and unchecked until signed; a reader elsewhere checks their own rules with the competent authorities before building, and nothing here replaces that.

1. Purpose

Water for the tree nursery from a roof: gutters and downpipes, a first-flush diverter on each, a buried cistern near the seedbeds, and a hand pump — no electricity, no treatment, and nothing taken from the spring.

Five rules. Two waters, kept apart: roof water for plants, spring water for people; no pipe, valve, or hose joins them. The first water is thrown away: every downpipe passes a first-flush diverter before the cistern. Never topped up from the spring: in a drought longer than the cistern, water is carried from the market garden's pond, which the stream refills. Covered and locked: the cistern's one manhole is locked, and the pump is the only way in. Close to the beds: the cistern sits low, beside the seedbeds, so the water is carried a few steps, not across the site.

Connections. Depends on: a roof of about 100 m² near the nursery, from the buildings modules; the tree nursery module, whose beds and watering it serves; the Phase 1 walk, with the roof, its downpipes, the fall, and the cistern site; the rainfall record of the nearest hydrometeorological station; the written

enquiries in Section 10. Feeds: the tree nursery. Takes from: the market garden's pond, in a long drought only. Hands back to the water module: the non-drinking uses, sized in Section 3.3, only if the spring measures between its thresholds.

Patterns. Grows from: not yet named. How the valley's households caught the water off their roofs, and what they watered with it, are questions for the elders. Feeds: not yet named. Resonates with: 104 Site Repair — the cistern on poorer ground beside the seedbeds; 25 Access to Water — roof water open to every hand at the pump, where the spring's is fenced; 172 Garden Growing Wild — the nursery raising the trees the regeneration strand plants. In tension with: 64 Pools and Streams — a buried cistern takes the roof's water out of sight; the downpipes, the diverters, and the overflow are where it stays visible.

2. Context of validity

Condition	Range this design holds for
Climate	[set] — mountain; watering in the warm months
Rainfall	800–1,000 mm a year, [assumed]
Roof	about 100 m ² , hard and sloped, [set]
Nursery	about 50 m ² of seedbeds, [set]
Group size	none: the nursery sets the size, not the people
Fall	roof above the cistern; the cistern beside the beds
Frost	the cistern below the frost line, as the potable one

Where the roof is small, the rain light, or the nursery larger, the cistern empties sooner and the pond carries more; Section 3 shows how to read it.

3. Sizing method

What you must measure yourself before using this module:

- **Your rainfall** — the annual and monthly record of the nearest hydrometeorological station, and the warm months' share of it.
- **Your roof** — its plan area, its material, and where its downpipes can fall, from the walk.
- **Your nursery** — its bed area and its watering, from the tree nursery module and, once it runs, from the watering record.

- **Your cistern site** — the soil and the fall, from the walk.

Our example: rainfall, 800 to 1,000 mm a year, [assumed]; roof_area, 100 m², [set]; runoff_factor, 0.8; nursery_area, 50 m², [set]; nursery_watering, 10 L a m² a week, [assumed]; first_flush, 0.5 L a m² of roof, [assumed]. Values from the site parameters file, held in the workshop.

3.1 The yield

One millimetre of rain on one square metre of roof is one litre.

Quantity	Calculation	Our example
A year	$100 \text{ m}^2 \times 0.8 \times 800\text{--}1,000 \text{ mm}$	64,000–80,000 L
A warm month	70–80 percent over six months	7,500–10,700 L
Rain to fill the cistern	$4,160 \text{ L} \div (100 \text{ m}^2 \times 0.8)$	about 52 mm

The rainfall. The paper's 800 to 1,000 mm is kept, [assumed]. A regional geography of the Ukrainian Carpathians gives 800 to 1,000 mm in the foothills and 800 to 1,200 mm at 800 to 1,000 m, rising to 1,400 to 1,600 mm on the highest peaks, with 70 to 80 percent of it in the warm season, mostly as downpours (Voropai and Kunytsia, Українські Карпати); the Encyclopedia of Modern Ukraine gives 500 to 800 mm in the foothills. At 500 mm the year's yield would be 40,000 L. The warm month above spreads the warm-season share over six months, April to September, [assumed]; the station record replaces it.

The runoff factor. 0.8, as the paper has it, now quoted: DIN 1989-1 gives a yield coefficient of 0.8 for a sloped hard roof, as reported by a German supplier's handbook (Amres); the Texas Manual on Rainwater Harvesting, 2005, p. 29, says most installers assume 75 to 90 percent, and notes that clay or concrete tile may lose up to 10 percent (p. 6).

3.2 The nursery and the cistern

Quantity	Calculation	Our example
A dry week, the paper's rate	$50 \text{ m}^2 \times 10 \text{ L}$	500 L
A dry week, at sowing	$50 \text{ m}^2 \times 14\text{--}17.5 \text{ L}$	700–875 L
A month, the paper's rate	$500 \text{ L} \times 4.3$	about 2,150 L
Cistern, internal	$2.0 \times 1.6 \text{ m} \times 1.3 \text{ m water}$	about 4.2 m ³
Dry weeks carried	$4,160 \text{ L} \div \text{the week}$	about 4.8 to 8.3

Quantity	Calculation	Our example
First flush, whole roof	$100 \text{ m}^2 \times 0.5 \text{ L}$	50 L

The nursery's need. The paper's 5 L a m^2 twice a week is kept, [assumed], for the season. It is below what forestry nursery guidance asks at sowing: the Ukrainian recommendations for growing seedlings of the main forest species give 10 L a m^2 for birch in open ground right after sowing, then every four to five days, about 14 to 17.5 L a m^2 a week (State Forestry Committee and the Ukrainian Research Institute of Forestry and Agroforestry, Kharkiv, 2010); the US Forest Nursery Manual gives 6.4 to 12.7 mm a day during germination on sandy soils (McDonald, in Duryea and Landis, 1984, ch. 12). The tree nursery module sets the rate, and the watering record replaces it.

The cistern. The paper's 4.2 m^3 , single chamber, the same construction as the potable cistern: 2.0 $\times$ 1.6 m inside, 1.3 m of water, below the frost line. It carries the nursery through about eight dry weeks at the paper's rate and about five at the sowing rate; the paper's six lies between. Over a year the roof gives several times what the nursery asks; the cistern, not the roof, is the limit, and about 52 mm of rain fills it from empty.

The first flush. The Texas Manual, p. 8, diverts at least 10 gallons for every 1,000 square feet of roof, about 0.41 L a m^2 , and reports an Australian study giving 0.53 to 2.0 L a m^2 . 0.5 L a m^2 is used, [assumed]: 50 L for the roof, shared among its downpipes. For two downpipes, [set], each diverter holds about 25 L — a standpipe of 160 mm pipe, with a bore of about 150 mm, [assumed], holding about 17.7 L a metre, some 1.4 m long, closed below by a cap with a small drip hole so that it empties itself between rains.

3.3 When the spring falls between its thresholds

The water module reads the spring against the demand: between 1.8 and 4 L a minute, the spring carries the people only if rainwater takes every non-drinking use it can. On this site that use is the flushing of the two existing flush toilets.

Quantity	Calculation	Our example
Flush water, a day	the greywater module, 3.1	about 283 L
Spring demand without it	$2.55 \text{ m}^3 - 0.28 \text{ m}^3$	about 2.27 m^3
As a flow	$2.27 \text{ m}^3 \div 1,440 \text{ min}$	about 1.6 L/min
One session's flushing	$283 \text{ L} \times 20 \text{ days}$	about 5.7 m^3
Rain to refill it, 100 m^2	$5,660 \text{ L} \div (100 \text{ m}^2 \times 0.8)$	about 71 mm

A session is session_length, 20 days, [assumed]. It takes its own roof and cistern, of about 6 m³, beside the flush toilets; a small pump from the energy module; and its own pipe to the toilets' cisterns, marked along its length, with no connection to the potable system. Frozen gutters and snow give little in winter, so a winter session's flushing falls back on the spring. It is sized here and not costed; it is designed only if the spring measures between the thresholds, and asked of the sanitary service first (Section 10).

4. Bill of materials

Quantities for the example. Alternatives are named in the design; none is yet known to work here.

Item	Quantity	Alternatives in the design
Gutters, downpipes, brackets	one 100 m ² roof	PVC; coated steel
First-flush diverter, 25 L	2	a bought downpipe filter
Cistern, masonry, 4.2 m ³	1	a 5,000 L plastic tank
Lockable manhole cover	1	none named
Inlet, overflow, washout, vent	1 set	none named
Hand pump, piston	1	a tap at the base of a tank
Watering cans and a short hose	[set]	a drip line

The cistern, as designed. As the potable cistern in the water module, Section 4, at about half its volume: one chamber, 2.0 × 1.6 m inside, 1.3 m of water and 0.3 m of freeboard; rubble stone walls in cement mortar; a reinforced floor and roof slab; at least 1 m of earth cover; one lockable manhole; an inlet through the diverters, an overflow led away downslope, a washout, and a screened vent.

The plastic tank, as the alternative elsewhere. Not chosen at this site: the household chose the masonry cistern on 2026-10-02. For a site that needs it: a 5,000 L tank above ground, opaque, on a level gravel base, drawn by gravity from a tap at its base, so that no pump is needed; emptied before the frost and left open to drain, [set]. It costs about half the masonry cistern and lasts [set]; a tank rated for burial was not found priced.

The siting, as designed. None has been measured yet; every rule below is checked on the ground before anything is dug.

From	To	Rule
The cistern and its pipes	the potable system	no connection
The overflow	any foundation	led away, [set]
The overflow	the spring's catchment	never into it
The cistern	septic tank, bed, soakaway	upslope of them
The seedbeds	the stream and a pond	25 m; 50 m on a slope
The cistern	the seedbeds	beside them, [set]

The seedbeds keep out of the Water Code's protective strip along the stream and around the pond, as the market garden's beds do (Section 10). No distance was found between a rainwater cistern and a septic tank or soakaway; the cistern stands above them on the fall, and the walk sets the rest.

The layout, as designed. Roof, gutters, downpipes, diverters, cistern, pump, seedbeds, in that order down the fall; the overflow away from every foundation and from the spring. Drawn as relationships only; the walk sets the rest.

5. Build

Step	Trade	Time	If done out of order
Enquiries answered	household	[from L2]	[from L2]
Walk: roof, fall, site	household	[from L2]	cistern misplaced
Excavation	excavator	[from L2]	[from L2]
Cistern walls, slabs, render	mason	[from L2]	[from L2]
Gutters, downpipes, diverters	roofer	[from L2]	first water in
Pump, overflow laid	plumber	[from L2]	overflow at footings
Backfill, earth cover	excavator	[from L2]	frost in the cistern
Filled by rain, pump tried	household	[from L2]	[from L2]

Built with the nursery, after the potable system, as the paper orders it.

6. Cost

Estimates, not quotations: each line is either a price listed by a Ukrainian retailer, seen on 2026-09-29, or the water paper's own estimate for autumn 2026, marked "paper". Place: not a quotation; western Ukrainian listings where found, national otherwise. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

6.1 Collection

Item	UAH	EUR	Basis
Gutters and downpipes, 100 m ² roof	12,000	231	paper
First-flush diverters, two	2,000	38	paper
Subtotal	14,000	269	

A 130 mm PVC gutter is listed at 360 to 462 UAH a 3 m length, a 100 mm downpipe at 565 UAH a 3 m length, and gutter brackets at 112 to 144 UAH each, in western Ukrainian listings; the roof's own lengths, from the walk, turn these into a quotation. No first-flush diverter was found in Ukrainian retail; it is built from pipe and fittings. A downpipe rainwater collector for a barrel, a different device, is listed at 351 to 380 UAH.

6.2 Storage and drawing

Item	UAH	EUR	Basis
Cistern, 4.2 m ³ , masonry	82,300	1,583	paper
Hand pump, piston, 8 m lift	3,500	67	estimated, listing
Subtotal	85,800	1,650	

The paper costs the cistern at 55 percent of the potable cistern's 149,630 UAH: about half the volume, one chamber, one manhole. The hand pump is a stainless piston pump listed at 3,500 UAH, against the paper's 5,000; the seller is not in western Ukraine.

6.3 Summary

Section	UAH	EUR
Collection	14,000	269

Section	UAH	EUR
Storage and drawing	85,800	1,650
Construction subtotal	99,800	1,919
Design and engineer's check, 8 percent	8,000	154
Contingency, 20 percent	20,000	385
Total external cost	127,800	2,458

Against the paper's 101,300 UAH before its percentages, the difference is the hand pump. The percentages are the paper's; the contingency is taken on the construction subtotal, as in the land and sanitation modules, where the water module takes it on the subtotal and the design together, as the paper does.

The plastic tank, not chosen here. For a site that would choose it: a 5,000 L tank is listed at 38,000 UAH, vertical, and 38,500 UAH, horizontal, in western Ukrainian listings; with the horizontal tank, and a tap, not priced, in place of the pump, the construction subtotal would be about 52,500 UAH, and the total about 67,200 UAH. The gravel base is not priced.

Contribution, not in the total. Rubble stone for the cistern from the site or the valley, with the landowner's leave, in proportion to the potable cistern's, [set]. The labour is in the cost.

Running cost. None bought. The gutters and diverters cleared in spring and autumn; the cistern inspected yearly and cleaned when silt shows; the pump's seals replaced when worn, [set]. The work is paid, [set] hours.

7. Operation and maintenance

Responsible: one named adult, paid, with a deputy — [from L2].

- **After heavy rain:** the diverters checked for draining; the overflow walked.
- **In the watering season:** the cistern's level read weekly and written down with the water drawn, so that the nursery_watering figure is replaced by a measured one.
- **Spring and autumn:** gutters and diverters cleared of leaves and needles.
- **Before the frost:** the pump drained; the diverters' caps opened; a plastic tank, where one is used instead, emptied.
- **Yearly:** the cistern inspected; cleaned when silt shows; the manhole's lock checked.

How the programme relates to this work is described in the programme modules, in standing format.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Cistern empty	level falls, no rain	water from the pond
Diverter blocked	it stays full	clear the drip hole
Gutter overflows	water down the wall	clear it; larger outlet
Pump frozen	no stroke in frost	drain it before frost
Overflow at footings	wet wall, soft ground	lead it further
Water green or smelling	colour at the pump	cover checked; clean
Lock or cover broken	cover loose	replace at once

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-09-29, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Read on 2026-09-29 on secondary copies, because the statute site could not be reached; each is read on zakon.rada.gov.ua and signed before this row counts as checked. No rule was found that speaks of collecting rain from one's own roof.

- **Water Code, No. 213/95-BP**, Art. 1: a water body is a natural or artificial element of the environment in which water collects — a sea, river, lake, pond, canal, and the like. Art. 47: general water use, from water bodies without structures or technical devices, needs no permit. Art. 48: special water use is taking water from water bodies with structures or technical devices; as quoted on a secondary copy, taking up to 5 m³ a day is not special water use. Our reading, not a finding: a roof and a cistern are not a water body, so the permit regime would not apply; the enquiry below asks.
- **Water Code, Arts. 88–89**: a protective strip of 25 m along small streams and around ponds, 50 m on a slope, in which gardening and ploughing are not allowed; the seedbeds stay out of it.

The written enquiries: to the regional office of the state water agency, whether roof water held in a cistern for watering is any kind of water use under the Code; to the council or planning authority, whether a buried cistern of 4.2 m³ needs notice or permission; and, only if Section 3.3 is built, to the district sanitary service, whether rainwater may flush toilets where children stay.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A drier valley**: a larger roof or a second one; a larger cistern; the pond or another source for the longest droughts.
- **A wetter valley**: the cistern stays at the nursery's need; the overflow is sized for the downpours.
- **A colder valley**: the cistern deeper under earth; the pump inside a small insulated housing, or drained all winter.
- **A larger nursery**: the cistern scales with the dry weeks it must carry; a drip line from a header tank saves water and work.
- **A smaller budget**: the plastic tank of Section 4, at about half the cost, emptied each winter.
- **No roof near the nursery**: a plastic tank under the nearest roof, and a hose, or the beds moved to the roof.
- **A tile roof**: runoff_factor lowered by up to a tenth.

12. Provenance and licence

- **Written by**: Michel Garand. [set — others as they choose to be named]

- **From whose knowledge:** the water working paper, EN v0.1; the Texas Manual on Rainwater Harvesting, 3rd ed., Texas Water Development Board, 2005; DIN 1989-1, as reported in a German supplier's handbook; Voropai and Kunytsia, Українські Карпати, on precipitation; the Encyclopedia of Modern Ukraine, entry Карпати; the Ukrainian recommendations for growing forest seedlings, Kharkiv, 2010; McDonald, in the Forest Nursery Manual, 1984; the Water Code of Ukraine as read on secondary copies; Ukrainian retail listings of 2026-09-29 for prices.
 - **Traditional knowledge:** none. How the valley caught roof water is for the elders.
 - **Text:** CC BY-SA 4.0
 - **Drawings and designs:** [hardware licence — open decision]
 - **Measurement data:** none yet
 - **Gate 1:** household decision, 2026-10-02, recorded by Michel Garand — v0.2 may be published as it stands, at L1, in English, Ukrainian, and German; v0.1 was published on 2026-09-29
 - **Wartime review:** 2026-10-02, Michel Garand — v0.2 read against compendium Section 2; nothing places the site; the rainfall is given at the resolution of the whole range; accepted
-

13. What to check

- **The enquiries** — water use, the cistern's standing, and, if built, flushing with rainwater.
 - **The rainfall** — annual and monthly, from the nearest hydrometeorological station, replacing the assumed range and the warm-season spread.
 - **The roof** — chosen, its plan area and material measured, its downpipes counted; if it is asbestos cement, a builder decides how the gutters are fixed without cutting or drilling it.
 - **The nursery** — nursery_area and nursery_watering from the tree nursery module, then from the watering record.
 - **The first flush** — first_flush tried against the water's colour after the first rains.
 - **The cistern** — masonry, chosen 2026-10-02; a mason's quotation replacing the paper's 55 percent.
 - **The gutters** — the listed prices turned into a quotation from the roof's lengths.
 - **The hand pump** — a seller in western Ukraine.
 - **The spring** — whether Section 3.3 is needed at all.
 - **The Ukrainian row** — every instrument read and signed.
 - **From L2, once built:** build times and out-of-order consequences in Section 5; the maintenance record in Section 7; early signs and repair costs in Section 8; Section 9.
-